

Chapter: 21

PHYSICS OF SOLIDS**PHYSICS OF SOLIDS:****Definition:**

"The Physics of solids is a branch of physics that deals with the study of the physical properties of solid materials, including their electronic, magnetic, optical, thermal and mechanical properties".

Key Features:

- It aims to understand the behavior of solid materials under different conditions, such as temperature, pressure and electromagnetic fields, using principles of quantum mechanics.
- It also helps to explain phenomenon like crystal structures, phase transitions, defects in material, and the interactions between electrons, ions and atoms in solids.

Example:

A key hurdle to enhance solar power which is an affordable energy source. It is increasing the efficiency of photovoltaic cells that convert solar radiation to electricity, making solid materials in solar cells as conductive as possible.

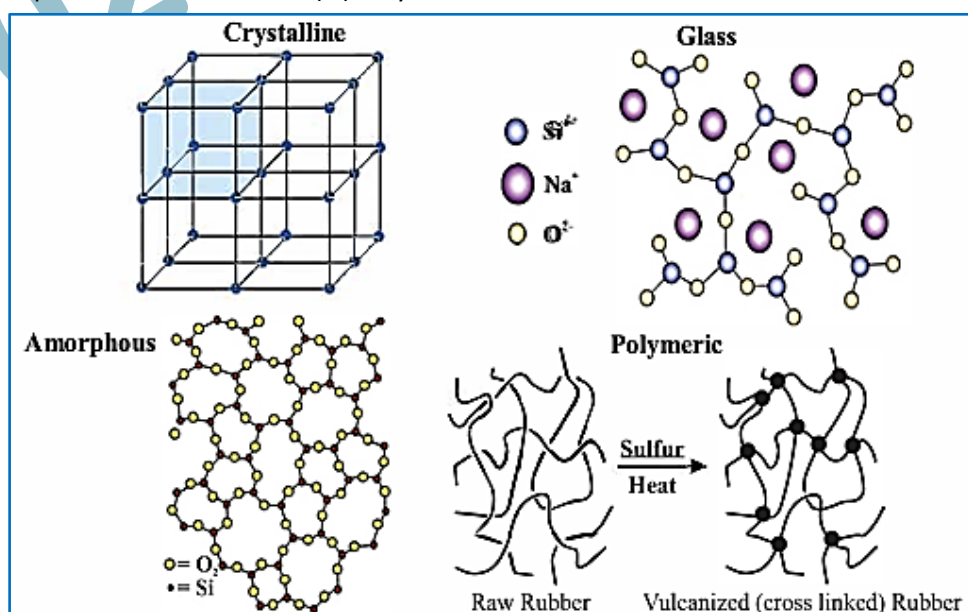
PIEZOELECTRIC EFFECT:

The Piezoelectric effect is the ability of certain materials to generate an electric charge in response to mechanical stress, such as pressure, vibration, or stretching, causing the material's internal electric dipoles to align and produce a voltage across the material, with the magnitude of the charge being directly proportional to the applied stress, allowing the material to convert mechanical energy into electrical energy.

CLASSIFICATION OF SOLIDS:

Solids are usually crystalline substances; their molecules are arranged in a definite pattern and in fixed positions. The way in which a solid behaves depends on its internal structure and there are four types of solid.

- | | |
|------------------------|------------------------|
| (i) Crystalline solids | (ii) Glassy solids |
| (ii) Amorphous solids | (iii) Polymeric solids |



21.1.1 Distinguish between structure of crystalline, glassy, amorphous and polymeric solids:

Property	Crystalline Solids	Glassy Solids	Amorphous Solids	Polymeric Solids
Atomic Arrangement	Regular, Repeating pattern.	Random, short-range order.	Random, no long-range order.	Long chains or networks of repeating units.
Melting Point	Sharp, Specific temperature.	Gradual, over a range of temperature.	Gradual, over a range of temperatures.	Varies, typically lower than crystalline solids.
Rigidity	Very rigid and well-defined structure.	Rigid, but less than crystalline solids.	Not as rigid as crystalline solids.	Can be flexible or rigid.
Transparency	Can be transparent or translucent.	Transparent or opaque depending on composition.	Opaque.	Depending on structures, can be transparent or opaque.
Examples	Diamond, salt crystals, silicon.	Window glass, certain plastics, amorphous metals.	Rubber, some plastics, glass.	Polyethylene, PVC, nylon.

DEFORMATION:

Definition:

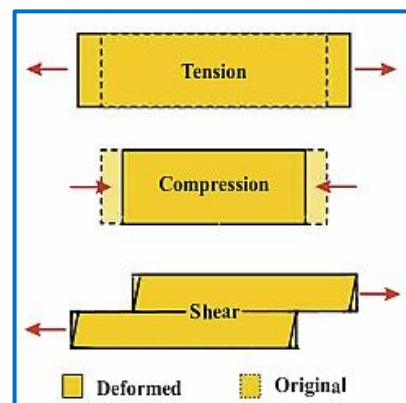
"The modifications of the shape or size of an object due to applied forces or a change in temperature is called deformation."

Explanation:

Deformation is usually caused by forces such as tensile (pulling) or Compressive (pushing) as shown in figure.

As deformations occur, internal inter-molecular forces arise that oppose the applied force.

Different types of deformations may result from variations in type of material, size and the forces applied.



DEFORMATION IN ONE DIMENSION (1D):

1. STRESS:

Stress is a quantity which defines the magnitude of force that causes deformation.

"The force applied to a material per unit area is called Stress."

Mathematically:
$$\text{Stress } (\sigma) = \frac{\text{Force } (F)}{\text{Area } (A)}$$

Unit: The SI unit of stress (σ) is newton per square metre (Nm^{-2}), which is given the name pascal (Pa).

TYPES OF STRESS:

(i) Tensile stress:

"The stress which increases the length of solid, is called Tensile Stress."

For example: When a rod or wire is stretched by pulling it with equal and opposite forces (outwards) at both ends.

(ii) Compressive stress:

"The stress which decreases the length of solid, is called Compressive Stress."

For example: When a rod or wire is compressed/squeezed by pushing it with equal and opposite forces (inwards) at both ends.

(iii) Shear stress:

"The stress which changes the shape of solid, it is called Shear Stress."

When the direction of the deforming force or external force is parallel to the cross-sectional area, the stress experienced by the object is called shearing stress or tangential stress. This results in a change in the shape of the body.

(iv) Bulk Stress (or Volume Stress):

"The stress which changes the volume is called Bulk Stress."

When the deforming force or applied force acts from all dimensions resulting in the change of volume of the object then such stress is called volumetric stress or Bulk stress.

2. STRAIN:

"Strain is the amount of deformation experienced by the body in the direction of force applied, divided by the initial dimensions of the body."

Strain is a measure of the deformation of material (stretches or compresses) when stress is applied to it.

The strain is a dimensionless quantity as it just defines the relative change in shape.

TYPES OF STRAIN:

Strain experienced by a body can be of four types depending on stress application as follows:

1. TENSILE STRAIN

If strain is due to tensile stress, it is called tensile strain.

The deformation or elongation of a solid body due to applying a tensile force or stress is known as Tensile strain. In other words, tensile strain is produced when a body increases in length as applied forces try to stretch it.

In the case of deformation in one dimension, strain is defined as the fractional change in length.

Mathematically: If ΔL is the change in length and L is the original length, then strain is given by

$$\text{Tensile Strain } (\epsilon) = \frac{\text{Change in length } (\Delta L)}{\text{Original Length } (L)}$$

Unit: Since strain is ratio of lengths, it is dimensionless, and therefore, has no units.

2. COMPRESSIVE STRAIN

If strain is produced because of compressive stress, it is termed as compressive strain.

Compressive strain is deformation in a solid due to the application of compressive stress.

In other words, compressive strain is produced when a body decreases in length when equal and opposite forces try to compress it.

3. VOLUMETRIC STRAIN:

"The change in volume per unit volume is known as volumetric strain."

$$\text{Volumetric Strain} = \frac{\text{Change in Volume } (\Delta V)}{\text{Original Volume } (V)}$$

4. SHEAR STRAIN:

When the opposite faces of a rigid cube are subjected to shear stress, the shear strain produced is given by

$$\text{Shear Strain } (\gamma) = \frac{\Delta x}{L} = \tan \theta$$

ELASTIC MODULUS:

Definition:

"The ratio of Stress to Strain is constant and is called modulus of elasticity."

Explanation:

In the linear limit of low stress values, the general relation between stress and strain is known as Hooke's law is given as under.

$$\text{Stress} = (\text{elastic modulus}) \times \text{Strain}$$

$$\text{Elastic Modulus} = \frac{\text{Stress}}{\text{Strain}}$$

The greater the stress, the greater the strain; however, the relation between strain and stress does not need to be linear.

It is clear from the above equation that:

- When an object has a large value of elastic modulus, the effect of stress is small.
- For a small elastic modulus, it means that stress produces large strain and small deformation occurs.

Example:

Stress on a rubber band produces larger strain (deformation) than the same stress on a steel band of the same dimensions because the elastic modulus for rubber is two orders of magnitude smaller than the elastic modulus for steel.

TYPES OF ELASTIC MODULUS:

1. YOUNG'S MODULUS (MODULUS OF ELASTICITY)

Definition:

"In the case of linear deformation, the ratio of tensile (or compressive) stress (σ) to tensile (or compressive) strain (ϵ) is called Young's modulus."

It is denoted by the symbol Y and has units of $Y = \text{Force} / \text{Area}$ (Pascal, Pa or N/m^2).

Explanation:

It is a mechanical property of solids that measures the stiffness or elasticity of a solid material, and it determines how much a material will deform (stretch or compress) under a given force.

It is also a measure of a material's resistance to elastic deformation when subjected to an external force. The higher the Young's Modulus, the stiffer the material.

Mathematically:

Hooke's law can be written for Young's Modulus is defined as the ratio of stress to strain.

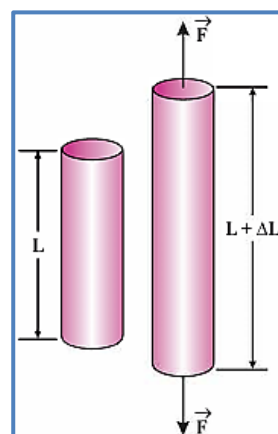
$$Y = \frac{\text{Stress}}{\text{Strain}}$$

$$Y = \frac{F/A}{\Delta L/L}$$

$$Y = \frac{F \times L}{\Delta L \times A}$$

Example:

Steel has a higher Young's Modulus than rubber, which makes steel more resistant to deformation than rubber when subjected to a force.



WORKED EXAMPLE:

- 21.1: Calculate the change in length of the upper leg bone (the femur) when a 70.0 kg man supports 62.0 kg of his mass on it, assuming the bone to be equivalent to a uniform rod that is 40.0 cm long and 2.0 cm in radius. Young's modulus for bones is $19 \times 10^9 \text{ N/m}^2$.

SELF-ASSESSMENT QUESTIONS:

1. Calculate the amount of stress in steel cable. Suspension cables are used to carry gondolas at Murree Hills. Consider a suspension cable that contains an unsupported span of 1.5 km with a diameter of 3.0 cm, with maximum tension it can withstand is $3.0 \times 10^6 \text{ N}$.

2. SHEAR MODULUS:**Definition:**

Shear modulus is the measure of the rigidity of the body which is the ratio of shear stress to shear strain.

It is denoted by the symbol G or S and has units of force per unit area, typically expressed in Pascal (Pa) or Giga Pascal (GPa).

Explanation:

Shear modulus, also known as the modulus of rigidity, is a mechanical property that describes a material's resistance to shear deformation when a force is applied perpendicular to the material's surface.

Mathematically:

Consider a rigid body as shown in figure. When acted upon by tangential force to twist it.

The shear stress is F/A and shear strain is $\Delta x/L$. Therefore, the shear modulus becomes.

$$G = \frac{\frac{F}{A}}{\frac{\Delta x}{L}}$$

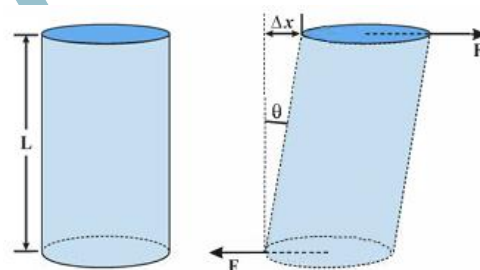
Where shearing Strain; $\frac{\Delta x}{l} = \tan \theta$

Therefore

$$G = \frac{\frac{F}{A}}{\tan \theta}$$

Hence

$$G = \frac{F}{A \cdot \tan \theta}$$

**WORK EXAMPLE:**

- 21.2: The area of the upper face of a rectangular block is 0.25 m^2 and the lower face is fixed. The height of the block is 1.5 cm, a shearing force applied to the top face which produces a displacement of 0.0152 mm. Find the strain, stress and the shearing force. The modulus of rigidity is $4.5 \times 10^{10} \text{ N/m}^2$.

3. BULK MODULUS:**Definition:**

Bulk modulus is defined as the ratio of the change in pressure applied to a material to the resulting fractional change in volume that occurs in the material.

It is denoted by the symbol K and has units of pressure (usually expressed in Pascal, Pa or Giga Pascal (GPa)).

Explanation:

For three-dimensional deformation, when volume is involved, then the ratio of applied stress to volumetric strain is called Bulk modulus.

Bulk modulus, also known as the modulus of compressibility, is a mechanical property that describes the resistance of a material to compression under uniform external pressure.

Mathematically:

Mathematically, it is expressed as: $B = \frac{\Delta P}{-\left(\frac{\Delta V}{V}\right)}$ Or $B = \frac{F/A}{-\left(\frac{\Delta V}{V}\right)}$

Where B is the bulk modulus, ΔP is the change in pressure applied to the material, ΔV is the resulting change in volume, and V is the original volume of the material.

FORCE- EXTENSION GRAPH:

In material science, the force versus extension graph for a material gives the relationship between stress and strain.

This is obtained by gradually applying a load to a test component and measuring its deformation according to tensile standards.

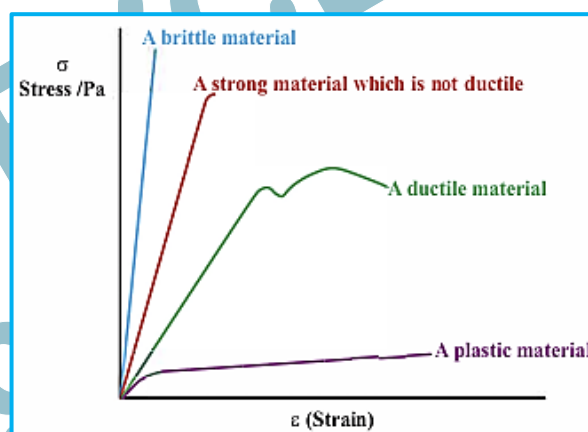
GRAPH OF DIFFERENT TYPES OF MATERIALS:

The curves obtained by graph between Strain and Stress reveal many properties of materials, such as the young's modulus, the yield strength, the ultimate tensile strength and so on, for different types of materials as shown in figure.

1. BRITTLE MATERIAL:

The blue line (brittle) represents the behavior of a typical elastic material, which follows Hooke's Law.

- As the material is stretched, the force applied to it increases linearly with the extension, up to a point called the yield point.
- The material eventually reaches a maximum point of extension, is called the ultimate extension or fracture point, beyond which it breaks or fractures.

**2. DUCTILE MATERIAL:**

The green line (ductile) represents the behavior of a typical viscoelastic material, which exhibits time-dependent deformation under a constant force.

In this case, the material initially undergoes elastic deformation, similar to the elastic material, but then continues to deform slowly over time under a constant force, eventually reaching a maximum extension or creep limit.

3. PLASTIC MATERIAL:

The purple line (Plastic) represents the behavior of a typical plastic material, which exhibits permanent deformation when subjected to a force.

In this case, the material undergoes plastic deformation immediately, with no linear region or yield point, and continues to deform permanently as the force is increased, until it ultimately fractures or breaks.

4. STRONG MATERIAL:

The Red line shows strong material, which is not ductile, like steel wires stretch very little and break suddenly.

5. POLYMERIC MATERIALS:

Polymer curves are produced by stretching a sample at a constant rate through the application of a tensile force. By using a constant rate of testing the strain-rate dependency of polymer behavior is not allowed to dominate. However, it should be appreciated that polymers have a marked inherent time - dependence in their response to deformation, which sets their behavior apart from other classes' materials.

STRESS-STRAIN CURVE:

Relationships between stress and strain can be plotted on a graph for most of the materials by an experiment for tensile stress. In this experiment, the force (stress) gradually increases, and it produces strain.

The values of stress and the strain are plotted on a graph. This graph is called the stress-strain curve.

REGIONS OF GRAPH:

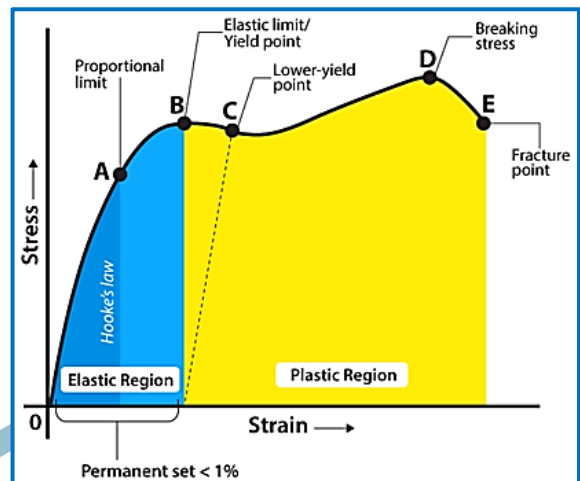
(i) Proportional Limit: OA

From point O to point A, the stress-strain curve is linear. Here, we can clearly see that Hooke's law is obeyed.

The body may regain its original dimensions when the applied force is removed.

In this limit, the stress-strain ratio gives a proportionality constant, known as Young's modulus and the slope of this line is the material's **Young's modulus**.

In this region, the solid behaves like an elastic body. Point A in the graph represents the proportional limit.



(ii) Elastic Limit: AB

From regions A to B, the stress and strain are not proportional, the graph doesn't obey Hooke's law. Up to this point B, the material will return to its original shape when the stress is removed. This point B is called the elastic limit.

Beyond this point, permanent deformation begins.

(iii) Yield Point: BC

Point B in the curve is known as the yield point, also known as the elastic limit in the stress-strain curve.

The corresponding stress applied to the object is known as the yield strength of the material.

If the object is stretched beyond the elastic limit B, it enters the plastic region and it does not return to the original length upon removal of the external force but remains permanently deformed, such is elastic deformation.

The yield point is defined as the point at which the material starts to deform plastically.

There are two yield points (i) upper yield point (ii) lower yield point. After the yield point is passed, permanent plastic deformation occurs.

(iv) Ultimate Stress Point: CD

It is a point that represents the maximum stress that a material can endure before failure. Beyond this point, failure occurs.

When the load is removed, at some point C, between B and D, the body does not regain its original dimension.

In such a case, even when the stress is zero, the strain is not zero.

The material is said to have a permanent set, and the deformation is known as plastic deformation.

Point D on the graph is known as the ultimate tensile strength.

Ultimate Tensile Strength: is a measure of the toughness of the material.

Ultimate Tensile Strength is the maximum amount of stress a material can handle before it breaks or fractures.

The maximum force that can be applied without breaking is called the ultimate tensile strength of the material.

It is measured in Pounds Per Square Inches (PSI).

(v) Plastic deformation:

When the stress is sufficient to permanently deform the metal (i.e., soft metal like silver, gold) it is called plastic deformation, and this region is called the Plastic Region.

Plastic Region: It is part of the Stress-Strain curve, where the material undergoes plastic deformation permanently and can't attain its initial state after the stress is removed.

Strain hardening modulus: The slope of the Stress-Strain Curve in the plastic region is called Strain hardening modulus and this modulus represents the ability of the material to resist further deformation.

(vi) Fracture or Breaking Point: E

It is the point in the stress-strain curve at which the failure of the material takes place or the point at which the material breaks.

The point in the Stress-Strain Curve, where material breaks in the experiment, is called the Fracture Point, and the stress or the force at this point is called Fracture strength.

ELECTRICAL PROPERTIES OF SOLIDS:

The electrical property of a good conductor depends on its ability to transmit energy without boiling, melting, or changing its composition in any way.

RESISTIVITY:

The resistance per unit length and cross-sectional area is called resistivity.

It varies from material to material. Its SI unit is Ohmmeter (Ω - m).

CONDUCTIVITY:

It is a measure of a material's ability to conduct electric current, which is influenced by the availability and mobility of charge carriers within the solid.

TEMPERATURE COEFFICIENT OF RESISTIVITY:

The temperature coefficient of resistance is defined as the change in electrical resistance of a substance with respect to per degree rise in temperature.

When the temperature increases, the process of electron collision becomes rapid and faster. As a result, the resistance will increase with the rise in temperature of the conductor.

THE QUANTITY OF CHARGE CARRIERS:

The number of charge carriers per unit volume can be found from measurement of the Hall Effect.

Its SI unit is m^{-3} .

ENERGY BANDS IN SOLIDS:**ENERGY BANDS:****Definition:**

"In solids, several bands of energy levels are formed due to the intermixing of atoms in solids. We call these sets of energy levels as energy bands".

Explanation:

In gaseous substances, the arrangement of molecules is spread apart and are not so close to each other. In liquids, the molecules are closer to each other.

But, in solids, the molecules are closely arranged together, due to these atoms of molecules tend to move into the orbitals of neighbouring atoms. Hence, the electron orbitals overlap when atoms come together.

When isolated atoms come together to form a solid, interactions between neighboring atoms cause the electron energy levels to split and overlap, creating continuous energy bands.

Electrons of an isolated atom are bound to the nucleus and can only have distinct energy levels. However, when a large number of atoms, say N , are brought close to one another to form a solid, each energy level of the isolated atom splits into N sub-levels, called states, under the action of the forces exerted by other atoms in the solid. These permissible energy states are discrete but so closely spaced that they appear to form a continuous energy band.

FORMATION OF ENERGY BANDS:

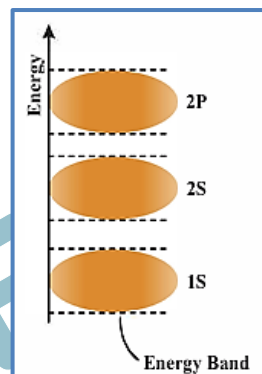
In an isolated atom, the electrons in each orbit possess definite energy. But, in the case of solids, the energy level of the outermost orbit electrons is affected by the neighbouring atoms.

When two isolated charges are brought close to each other, the electrons in the outermost orbit experience an attractive force from the nearest or neighbouring atomic nucleus.

Due to this reason, the energy levels of the electrons will not be at the same level, the energy levels of electrons are changed to a value which is higher or lower than that of the original energy level of the electron.

The electrons in the same orbit exhibit different energy levels. The grouping of these different energy levels is known as the energy band.

However, the energy of the inner orbit electrons is not much affected by the presence of neighbouring atoms.



CLASSIFICATION OF ENERGY BANDS:

1. VALENCE BAND:

Definition:

The electrons in the outermost shell are known as valence electrons. These valence electrons contain a series of energy levels and form an energy band known as the valence band.

Explanation:

The valence band has the highest occupied energy and contains electrons under normal conditions. It may be either completely filled or partially filled with electrons and can never be empty.

The valence band is the range of energy levels occupied by electrons that are bound to atoms and participate in chemical bonding.

2. FORBIDDEN BAND (ENERGY GAP):

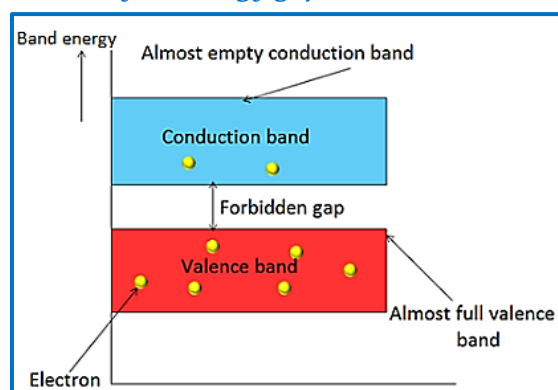
Definition:

"The valence band and the conduction band are separated by an energy gap known as the forbidden band."

Explanation:

The valence band contains the lower energy states, while the conduction band contains higher energy states. This energy gap prevents electrons in the valence band from easily moving to the conduction band, significantly influencing the electrical properties of the solid, such as its ability to conduct electricity.

In between two consecutive permissible energy bands, there is a range of energy states which cannot be occupied by electrons. These are called forbidden energy states, and their



range is termed as forbidden energy gap. The gap between the valence band and the conduction band is referred to as the forbidden gap.

As the name suggests, the forbidden gap doesn't have any energy, and no electrons stay in this band.

If the forbidden energy gap is greater, then the valence band electrons are tightly bound or firmly attached to the nucleus. We require some amount of external energy that is equal to the forbidden energy gap.

This gap influences the material's electrical properties; a larger gap typically means the material is less conductive.

3. CONDUCTION BAND:

Definition:

The conduction band is the range of energy levels where electrons are free to move throughout the material, allowing electrical conduction.

Explanation:

The valence electrons are not tightly held to the nucleus due to which a few of these valence electrons leave the outermost orbit even at room temperature and become free electrons.

The free electrons conduct current in conductors and are therefore known as conduction electrons.

The conduction band is one that contains conduction electrons and has the lowest occupied energy levels.

Electrons in the conduction band are not bound to any particular atom and can carry electric current through the solid.

The band above the valence band is called conduction band.

In the conduction band, electrons move freely and conduct electric current through solids. That is why the electrons occupying this band are known as conductive electrons or free electrons.

Any electron leaving the valence band is accommodated by this band. It may be either empty or partially filled with electrons.

In the case of a diamond crystal, carbon atoms with the electronic structure $1s^2 2s^2 2p^2$ form two main energy bands as shown in figure.

ELECTRICAL CLASSIFICATION OF SOLIDS:

Depending on the electrical conductivity, solids are classified into three main categories:

1. Insulators.
2. Conductors.
3. Semiconductors

1. INSULATORS:

Insulators are those materials in which valence electrons are bound very tightly to their atoms and are not free.

In terms of energy bands, it means that an insulator has

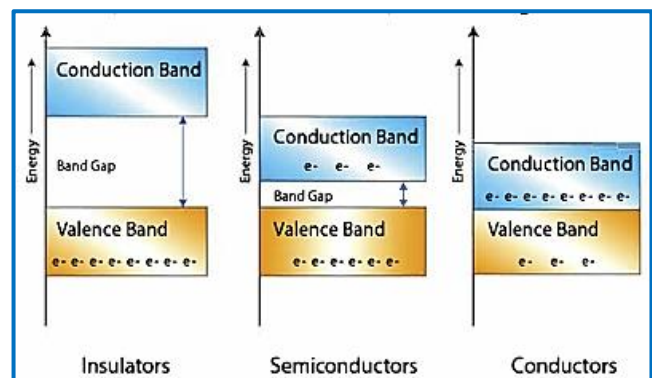
1. An empty conduction band (no free electrons)
2. A full valence band
3. A large energy gap (several eV) between them.

These materials do not conduct electricity. The band gap between the valence band and conduction band is very large. Even if a large amount of energy is provided to these solids, they do not conduct electricity, such as wood, plastics etc.

2. CONDUCTOR:

Conductors are those which have plenty of free electrons for electrical conduction.

In terms of energy bands:



Conductors are those materials in which valence and conduction bands largely overlap each other.

There is no gap between the conduction band and the valence band. Thus, electrons can easily flow from the valence band to the conduction band under the influence of an electric field, making them good conductors of electricity.

3. SEMICONDUCTORS:

Semiconductors are materials that have a conductivity between conductors and insulators.

In terms of energy bands, semiconductors are the materials which at room temperature have:

1. Partially filled conduction band
2. Partially filled valence band
3. A very narrow forbidden energy gap (of the order of 1 eV) between the conduction and valence bands.

The gap between the conduction band and the valence band is very small; therefore, whenever sufficient energy is provided to the electrons in a semiconductor, electrons jump from the valence band to the conduction band.

EFFECT OF TEMPERATURE ON SEMICONDUCTORS:

The conductivity of the semiconductors increases with an increase in temperature, for semiconductors, it lies in between 10^{-6} to $10^{-4} (\Omega m)^{-1}$.

At 0K, there are no electrons in the conduction band, and their valence band are completely filled. It means at 0K, a piece of Ge or Si is a perfect insulator.

However, with the increase in temperature, some electrons possess sufficient energy to jump across the small energy gap from valence to conduction band.

This transfers some free electrons in the conduction bands and creates some vacancies of electrons in the valence band.

The vacancy of electron in the valence band is known as a hole.

It behaves like a positive charge. Thus, at room temperature, Ge or Si crystal becomes a semiconductor.

TYPES OF SEMICONDUCTORS:

There are two types of semiconductors:

1. INTRINSIC SEMICONDUCTOR:

Definition:

A semi-conductor in its extremely pure form is known as intrinsic semi-conductor.

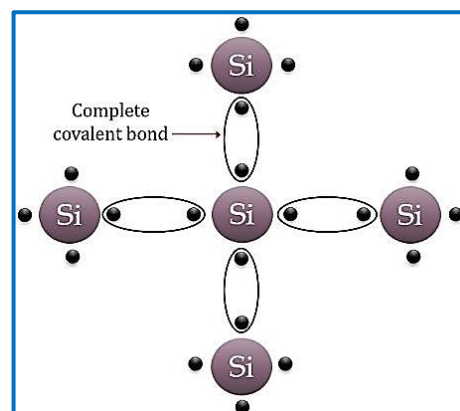
Pure elements of silicon and germanium are intrinsic

Explanation:

These semi-conductor elements have atoms with four valence electrons.

In solid crystalline form, the atoms of these elements arrange themselves in such a pattern that each atom has four equidistant neighbours.

Figure shows this pattern along with its valence electrons. Each atom with its four valence electrons, shares an electron from its neighbours. This effectively allocates eight electrons in the outermost shell of each atom which is a stable state. This sharing of electrons between two atoms creates covalent bonds. Due to these covalent bonds' electrons are bound in their respective shells.



When energy is provided to intrinsic semiconductors like silicon and germanium, electrons leave their positions, creating positive holes.

Under an electric field, electrons and holes move in opposite directions, enabling electrical conductivity.

Group IV elements are considered as intrinsic semiconductors.

2. EXTRINSIC SEMICONDUCTOR:

Definition:

The doped semi-conducting materials are called extrinsic semi-conductors.

Explanation:

These are semiconductors that have been doped with impurities to modify their electrical properties.

The electrical behaviour of semiconductors is extremely sensitive to the purity of the material. It is substantially changed on introducing a small Si amount of impurity into the pure semi-conductor lattice.

The process is called doping, in which a small number of atoms or some other suitable elements are added as impurity in the ratio of 1 to 10^6 .

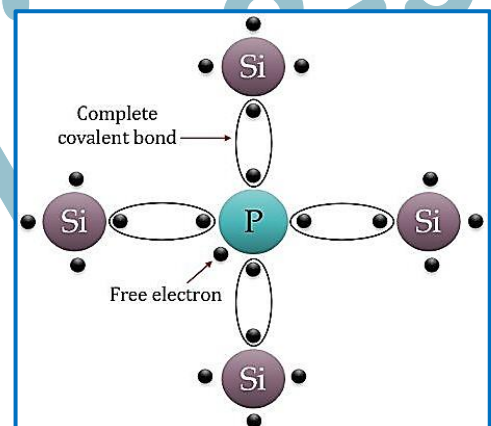
Silicon and Germanium have low conductivity at room temperature. Adding small amounts of group III or group V impurities, a process called doping, increases their conductivity, resulting in extrinsic semiconductors.

TYPES OF EXTRINSIC SEMICONDUCTORS:

n-type semi-conductor:

When a silicon crystal doped with a pentavalent element, e.g., arsenic, antimony or phosphorous etc., four valence electrons of the impurity atom form covalent bond with the four neighbouring Si atoms, while the fifth valence electron provides a free electron in the crystal. Such a doped or extrinsic semi-conductor is called n-type semi-conductor.

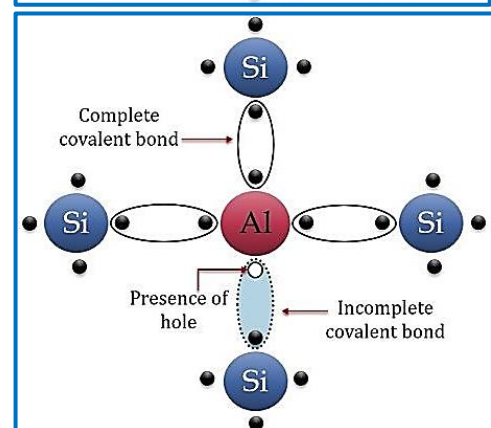
Figure illustrates silicon crystal lattice doping with a pentavalent impurity such as phosphorous. The phosphorous atom is called a donor atom because it readily donates a free electron, which is thermally excited into the conduction band.



p-type semi-conductor:

When a silicon crystal is doped with a trivalent element, e.g., aluminium, boron, gallium or indium etc., three valence electrons of the impurity atom form covalent bond with the three neighbouring Si atoms, while the one missing electron in the covalent bond with the fourth neighbouring Si atom, is called a hole which in fact is vacancy where an electron can be accommodated. Such a semiconductor is called p-type semi-conductor.

Figure illustrates silicon crystal lattice doped with aluminium. The aluminium atom is called an acceptor atom because it is easy for the aluminium ion core to accept a valence electron from a nearby silicon atom, thus creating a hole in the valence bond.



SUPERCONDUCTOR:

Definition:

A superconductor is a material that can conduct electricity without resistance when it is cooled below a certain critical temperature, typically near absolute zero.

Explanation:

There are some materials whose resistivity becomes zero below a certain temperature T_c (called critical temperature). Below this temperature, such materials are called superconductors.

They offer no resistance to electric current and are, therefore, perfect conductors.

Once the resistance of a material drops to zero, no energy is dissipated and the current, once established, continues to exist indefinitely without the source of an emf.

Discovery:

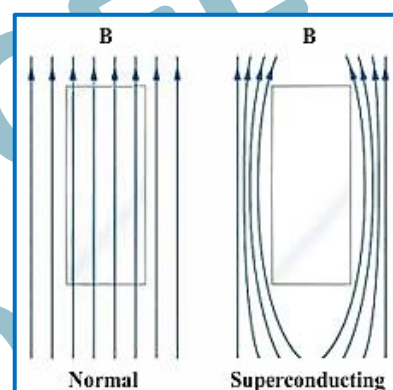
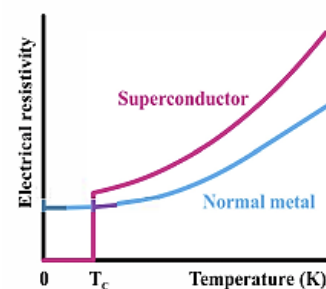
The first superconductor was discovered in 1911 by Kamerlingh Onnes when it was observed that electrical resistance of mercury disappears suddenly as the temperature is reduced below 4.2 K.

In 1986 a new class of ceramic materials was discovered that they become superconductors at temperatures as high as 125 K.

Any superconductor with a critical temperature above 77 K (the boiling point of liquid nitrogen) is referred to as a high temperature superconductor.

Near absolute zero, this allows charge to flow without energy loss to thermal effects. Superconductors exhibit this behavior below a critical temperature, usually near absolute zero, with resistivity less than $4 \times 10^{-25} \Omega\text{-m}$.

They also exhibit perfect diamagnetism, expelling magnetic fields in the superconducting state, known as the Meissner effect, and enabling phenomena like magnetic levitation.

**SUPER CONDUCTOR APPLICATIONS:**

Superconductors are used in particle zero accelerators, generators, transportation, computing, electric motors, medical applications, and power transmission, among other fields.

1. MAGNETIC RESONANCE IMAGING (MRI):

MRI scanners provide high resolution pictures of the tissues inside the body.

MRI is a diagnostic imaging technique used in radiology to display more information about the internal body structures clearer than X-Ray.

It can be used to create representations of the anatomy and physiological functions of the body in both illness and health.

HOW DOES MRI WORK?

Most of the magnetic resonance imaging (MRI) scanners have the form of large tubes.

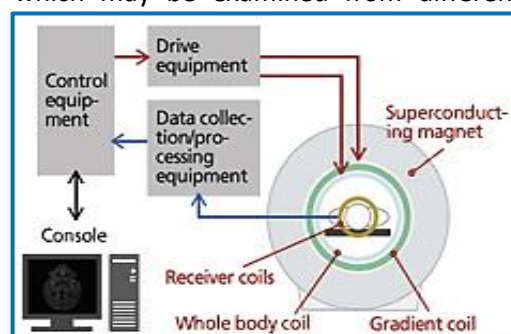
When a patient lies down inside an MRI scanner, the magnetic field causes the water molecules in your body to rearrange themselves temporarily.

These aligned atoms emit tiny signals when exposed to radio waves, which are then utilized to make cross-sectional MRI pictures, which are analogous to slices cut through a loaf of bread.

Also, the MRI device can produce three-dimensional pictures, which may be examined from different perspectives.

Explanation:

- Superconducting coils produce a strong magnetic field (up to 60,000 times as strong as the intensity of Earth's magnetic field) that is used to align the protons of hydrogen atoms in the body of the patient, as shown in figure.



- Like electrons, protons have a spin property, so they align with a magnetic field. The proton's axis vibrates about the applied magnetic field. Vibrating protons are crushed with a burst of radio waves tuned to push the Protons' spin axes sideways, perpendicular to the applied magnetic field.
- When radio waves pass and the protons quickly return to their vibrating pattern, they emit a faint electromagnetic signal whose frequencies depend slightly on the chemical environment in which the proton resides.
- These signals, detected by sensors, are then analyzed by a computer to reveal varying densities of hydrogen atoms in the body and their interactions with surrounding tissue.
- The resulting images clearly distinguish between fluid and bone. MRI was formerly called NMRI (Nuclear Magnetic Resonance Imaging) because hydrogen nuclei resonate with the applied fields.

USES:

1. MRI is used in medical applications. MRI accurately detects health problems and illnesses in the human body.
2. MRI can be used to treat brain tumors, heart attacks and strokes, head or neck injuries, brain and spinal cord defects, joints and bones, liver and other stomach problems, women's uterine irregularities, etc.

2. MAGLEV TRAIN SYSTEMS:

Maglev Trains are trains that use magnetic levitation to move without touching the ground. This is done by magnets which both lift the train up from the ground and propel the train forward.

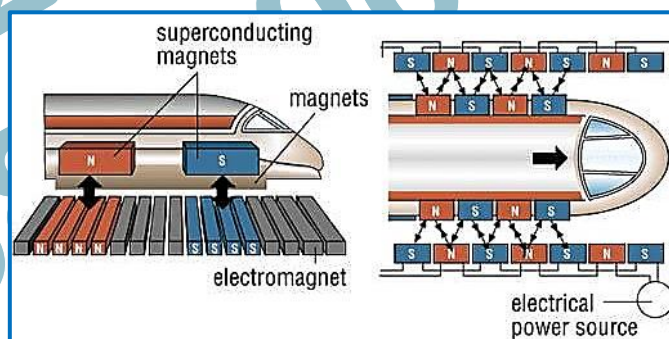
Maglev train systems use powerful electromagnets to float the trains over a guide way, instead of the old steel wheel and track system as shown in figure.

Construction & Working:

A system called electromagnetic suspension suspends, guides, and propels the trains.

A large number of magnets provide controlled tension for lift and propulsion along a track.

Maglev (derived from magnetic levitation) is the aim of train transportation that uses two sets of magnets.



1. One set to repel and push the train up off track.
2. Another set to move the elevated train ahead, taking advantage of the lack of friction.

3. SMALL ELECTRIC MOTOR:

Precision Instruments:

Superconducting motors can be used in precision instruments where high efficiency and low noise are critical.

Medical Devices:

Compact, efficient motors for medical devices such as MRI machines, which already use superconducting magnets.

Aerospace:

High-efficiency, lightweight motors for aerospace applications, where every gram of weight saved is crucial.

Robotics:

Small, powerful motors for advanced robotic applications requiring precise control and high-power output.

4. SUPER CONDUCTIVITY AND SUPER COMPUTERS:

Super conducting components can improve the performance of super computers in several ways such as:

1. Super conducting qubits:

These are the basic building blocks of quantum computers, which are able to perform calculations at speeds that are exponentially faster than classical computers.

Superconducting qubits can maintain their quantum states for long periods of time, allowing for complex computations to be performed.

2. Faster processing speeds:

Super conducting components can process information at much faster speeds than traditional electronic components, resulting in faster computing times and the ability to handle larger datasets.

3. Lower power consumption:

Super conducting circuits require less power to operate than traditional electronic circuits, reducing energy costs and making them more environmentally friendly.

4. Improved reliability:

Superconducting components are less prone to errors and can operate for longer periods of time without failure, resulting in more reliable and efficient supercomputers.

However, the use of superconducting components in super computers is still in the experimental stage and faces several challenges.

MAGNETIC PROPERTIES OF SOLIDS:

The magnetic properties of solids depend on how their electrons behave in response to an external magnetic field.

TYPES OF MAGNETIC MATERIALS:

There are three main types of magnetic materials:

1. Ferromagnetic Materials
2. Paramagnetic Materials
3. Diamagnetic Materials

1. FERROMAGNETIC MATERIALS:

Definition:

Ferromagnetic materials are substances that exhibit strong magnetic properties due to the alignment of magnetic moments within the material.

These materials can retain their magnetization even after an external magnetic field is removed, making them essential in various technological and industrial applications.

Explanation:

These materials can retain their magnetization even after an external magnetic field is removed, making them essential in various technological and industrial applications.

Ferromagnetic materials have domains (small regions where the magnetic moments are aligned). When an external magnetic field is applied, these domains align with the field, resulting in strong magnetization.

These materials are integral to many devices, from everyday items like refrigerator magnets to advanced technologies like MRI machines.

Examples: Iron, Cobalt, Nickel.

2. PARAMAGNETIC MATERIALS:

Definition:

Paramagnetic materials are substances that are weakly attracted by an external magnetic field and do not retain the magnetic properties once the external field is removed.

Explanation:

Unlike ferromagnetic materials, the magnetic moments in paramagnetic materials are randomly aligned when there is no external magnetic field. However, in the presence of an external magnetic field, the magnetic moments tend to align with the field, resulting in a weak magnetization.

Paramagnetic materials have important applications in various fields, including medical imaging (MRI contrast agents), magnetic cooling, and quantum computing.

Examples: Aluminum, Platinum, Titanium.

3. DIAMAGNETIC MATERIALS:

Definition:

Diamagnetic materials are substances that create an induced magnetic field in a direction opposite to an externally applied magnetic field.

Explanation:

Unlike ferromagnetic and paramagnetic materials, diamagnetic materials are weakly repelled by a magnetic field. This effect is very weak compared to the effects seen in paramagnetic and ferromagnetic materials.

It means that Diamagnetic Materials have no intrinsic magnetic moment. When exposed to an external magnetic field, they develop a weak, negative magnetization that opposes the applied field.

Examples: Diamagnetic, Bismuth.

TYPES OF FERROMAGNETIC SUBSTANCES:

Ferromagnetic substances can be classified into two main categories based on their magnetic properties: Soft ferromagnetic substances and Hard ferromagnetic substances.

1. HARD FERROMAGNETIC:

Definition:

The type of ferromagnetic substances that retain their magnetization after an external magnetic field is removed.

Hard ferromagnetic materials, also known as permanent magnets.

Explanation:

These materials are characterized by their high coercivity and remanence. In other words, they require a strong external magnetic field to become magnetized and maintain a significant level of magnetization even when the external field is no longer applied.

Examples:

Materials like neodymium, samarium cobalt, and alnico are permanent magnets with high coercivity, suitable for stable magnetic fields in applications like speakers and generators.

SOFT FERROMAGNETIC:

Definition:

The type of ferromagnetic substances that require a small magnetic field to become magnetized and lose their magnetization quickly once the field is removed.

Soft ferromagnetic materials are easy to magnetize and demagnetize.

Explanation:

Soft ferromagnetic materials, also known as soft magnetic materials, are characterized by their low coercivity and high permeability.

They have low coercivity, which means they require a small magnetic field to become magnetized and lose their magnetization quickly once the field is removed.

These materials can easily become magnetized and demagnetized, making them ideal for applications where rapid changes in magnetization are required.

Examples:

Including iron, nickel, and cobalt, these materials have low coercivity, making them easy to magnetize and demagnetize.

They are used in devices like transformers and inductors that require rapidly changing magnetic fields.

MAGNETIC DOMAIN:

Definition:

A magnetic domain is a region within a ferromagnetic material where the magnetic moments of atoms are aligned in the same direction.

Magnetic Domain Theory

Magnetic Domain is a small section of material in which the magnetic fields of atoms add up to create a strong, uniform magnetic field.

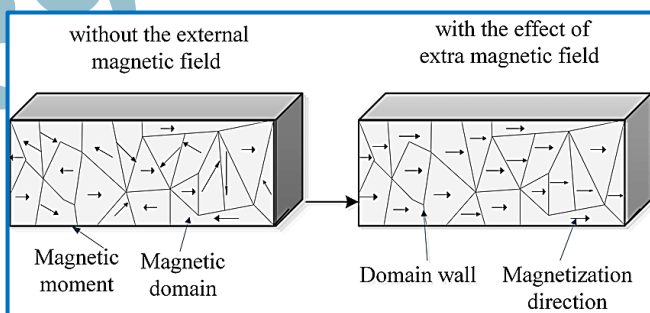
The magnetic domain theory was proposed by Weiss in 1907.

According to this theory:

Ferromagnetic material contains a large number of tiny regions, and each region (magnetic domain) consists of atomic magnetic moments that are aligned in a specific direction.

Initially, these magnetic domains are randomly oriented, resulting in a weak magnetization where the magnetic moments cancel each other out in the absence of an external magnetic field.

In the presence of a strong external magnetic field, these magnetic domains align in the direction of the field, resulting in strong magnetization.



This alignment occurs because the magnetic moments within each domain are parallel due to the influence of the external magnetic field. Paramagnetic, Ferromagnetic and Anti ferromagnetic as shown in Figure.

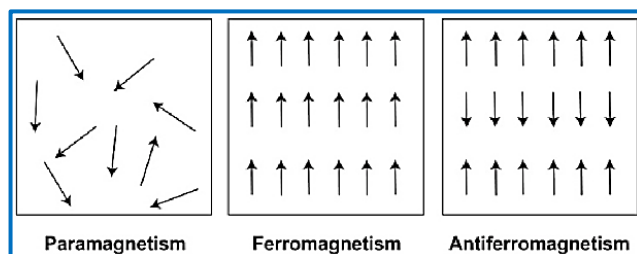
(a) Paramagnetic (Randomly oriented order)

(b) Ferromagnetic (Perfect order)

(c) Anti ferromagnetic (Perfect anti order)

In ferromagnetic materials, once all the magnetic domains are aligned in one direction and the external field is removed, the domain boundaries do not change their orientation. This property prevents demagnetization, making ferromagnetic materials useful for making permanent magnets.

Ferromagnetic materials are strongly attracted to external magnetic fields due to their aligned magnetic domains.



CURIE'S POINT AND CURIE'S LAW:

CURIE'S LAW:

The magnetization of paramagnetic materials is proportional to the applied magnetic field and inversely proportional to the temperature.

Statement:

This behavior is described by Curie's Law, which states that:

"The magnetization of a paramagnetic material is directly proportional to the applied magnetic field and inversely proportional to the absolute temperature".

Mathematically:

Mathematically, Curie's Law can be expressed as:

$$M \propto H \quad (H \text{ is the applied magnetic field strength})$$

But when it is heated, the relation is reversed, i.e., the magnetization becomes inversely proportional to temperature.

$$M \propto \frac{1}{T}$$

On combining both parts of statement, we get

$$M = \frac{C \chi H}{T} \dots\dots\dots (1)$$

Where: M is the magnetization; H is the applied magnetic field, T is the absolute temperature, and C is the Curie constant specific to the material.

The law implies that as the temperature increases, the magnetization of paramagnetic material decreases. Conversely, as the magnetic field increases, magnetization also increases.

This relationship is important for understanding the magnetic behavior of paramagnetic materials and their applications in various fields.

The various experiments by Pierre Curie showed that for many substances, the susceptibility (χ , chi) is inversely proportional to the absolute temperature T.

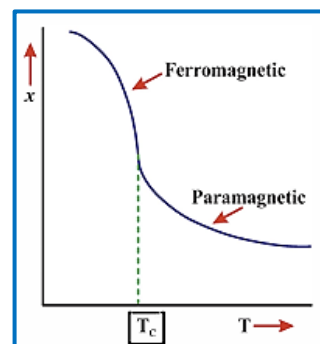
$$\chi = \frac{C}{T} \dots\dots\dots (2)$$

Example:

Consider an example of iron atoms with a temperature of 770°C (1043 K). Each iron atom acts like a tiny magnet at this temperature spontaneously. Each of them will align themselves as some kind of magnetic material.

For pure iron, as shown in Figure, atomic magnets are distributed within each microscopic domain. Pure iron is a kind of ferromagnetic material.

The directions of the magnetic fields are the same so that their magnetic fields strengthen each other.



ANTI-FERROMAGNETIC MATERIALS:

Anti-ferromagnetic materials possess atomic magnets that have alternate properties of magnetic fields.

They act in opposite directions so that their magnetic fields cancel each other.

CURIE POINT:

Pierre Curie, a French physicist, discovered it in 1895 and proposed certain laws related to the magnetic properties of materials as temperature changes.

Definition:

The Curie Point, or Curie Temperature, is the temperature at which certain materials lose their permanent magnetic properties.

Explanation:

Curie point is also known as Curie temperature. Some magnetic materials lose their magnetic properties as the temperature rises above the Curie temperature. Curie temperature weakens the magnetic properties of the material.

This point marks a critical transition where ferromagnetic materials (like iron, nickel, and cobalt) change into a paramagnetic state.

In simpler terms, the materials become magnetically disordered and can no longer retain a permanent magnetization.

Example:

Iron has a Curie Point of about 770°C (1420°F). When heated above this temperature, iron loses its ferromagnetic properties.

HYSTERESIS:

The meaning of hysteresis is “lagging”.

Definition:

Hysteresis is characterized as a lag of magnetic flux density (B) behind the magnetic field strength (H).

All ferromagnetic materials exhibit the phenomena of hysteresis.

Explanation:

Consider, a ferromagnetic substance is placed inside a current-carrying coil. Due to the magnetic field that is present, the substance gets magnetized. If we reverse the direction of the current, then the substance gets demagnetized, and this process is known as hysteresis.

HYSTERESIS LOOP:

A hysteresis loop represents the relationship between the magnetic field strength (H) and the magnetic flux density (B) of a ferromagnetic material.

Definition:

The hysteresis loop shows the relationship between the magnetic flux density and the magnetizing field strength.

Explanation:

This is a graphical representation of the relationship between the magnetic flux density (B) and the magnetic field strength (H) of a material.

It shows how the material responds to changes in the applied magnetic field and provides insight into its magnetic properties.

The loop is generated by measuring the magnetic flux coming out from the ferromagnetic substance while changing the external magnetizing field.

ENERGY LOSS DUE TO HYSTERESIS

During the cycle of magnetization and demagnetization of magnetic substances, energy is spent, and this spent energy appears in the form of heat. This heat loss is known as hysteresis loss.

The loss of energy per unit volume of the substance is equal to the area of the hysteresis curve.

Example:

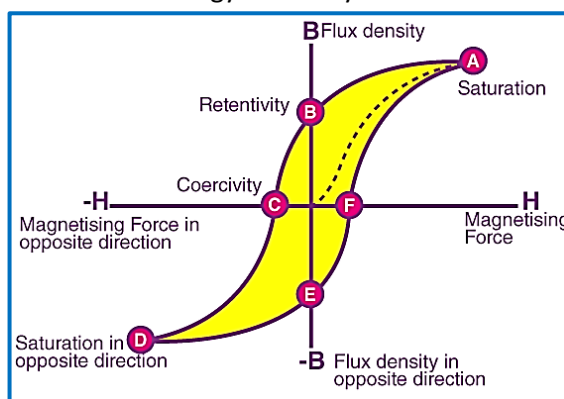
A transformer is the best example of studying energy loss due to hysteresis, as we know that during the process of magnetization and demagnetization, energy is required.

In transformers, due to the continuous process of magnetization and demagnetization, energy is lost in the form of heat continuously; due to this, the energy loss efficiency of the transformer gets reduced.

To stop this energy loss, a soft iron core is used in transformers because the energy loss or hysteresis loss in the case of soft iron is much smaller than in other materials.

UNDERSTANDING THE HYSTERESIS LOOP:

1. The magnetic flux density (B) increases when the magnetic field strength (H) increases from 0 (zero).
2. With an increase in the magnetic field, there is an increase in the value of magnetism, and it finally reaches point A, which is called the saturation point where B is constant.
3. With a decrease in the value of the magnetic field, there is a decrease in the value of magnetism. But if H is equal to zero, when a substance or material retains some amount of magnetism, it is called retentivity or residual magnetism.
4. When there is a decrease in the magnetic field towards the negative side, magnetism also decreases. At point C, the substance is completely demagnetized. The force required to remove the retentivity of the material is known as Coercive force (C).
5. In the opposite direction, the cycle is continued where the saturation point is D, the retentivity point is E, and the coercive force is F.
6. Due to the forward and opposite direction process, the cycle is complete, and this cycle is called the hysteresis loop.



RETENTIVITY AND COERCIVITY:

When a ferromagnetic material is magnetized by applying the external magnetizing field, after magnetization, if we remove the external magnetizing field, the material will not relax back to its zero-magnetization position.

RETENTIVITY:

The amount of magnetization present when the external magnetizing field is removed is known as retentivity.

It is a material's ability to retain a certain amount of magnetic property while an external magnetizing field is removed.

The value of the magnetic flux density (B) at point B in the hysteresis loop.

COERCIVITY:

The amount of reverse (-ive) external magnetizing field (H) required to completely demagnetize the substance is known as the coercivity of the substance.

The value of the magnetic field strength (H) at point C in the hysteresis loop.

ANALYZING THE LOOP:

The width of the hysteresis loop indicates energy loss (hysteresis loss) during magnetization and demagnetization cycles.

The shape of the loop provides information about the material's magnetic properties, such as coercivity and remanence.

By analyzing the hysteresis loop, the relationship between magnetic field strength and magnetizing current can be observed, aiding in understanding the material's magnetic properties and response to external magnetic field changes.

HYSTERESIS LOOP FOR PERMANENT AND TEMPORARY MAGNETS:

Hysteresis is especially marked in materials of high residual magnetism, such as hardened steel.

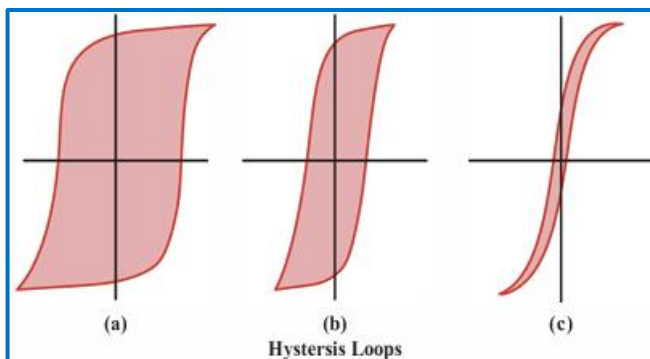
In most cases, hysteresis is a necessity as it causes dissipation of heat, waste of energy, and vibration due to change in polarity and rotation of element magnets in the material.

Hysteresis loops for hard steel, wrought iron, cast steel, and for alloyed sheet steel are shown in Figure.

1. Hysteresis loop for Hard steel:

Loop(a) is for hard steel. Due to its high retentivity power and large coercive force, this material is well-suited for permanent magnets.

Since the area of the hysteresis loop for hard steel is large, hard steel is not suitable for rapid reversals of magnetization.

**2. Hysteresis loop for wrought iron and cast steel:**

Loop (b) is for wrought iron and cast steel, which rise steeply. Hence, these materials have high magnetic permeability and good retentivity; therefore, these materials are suitable for cores of electromagnets.

3. Hysteresis loop for iron, low carbon steel & alloys:

Loop(c) is for iron, low carbon steel, silicon alloys, permalloy (alloy of Nickel and Iron) or Mu-metal (Nickel-Iron soft ferromagnetic material) sheets.

Since the permeability of these materials is very high and hysteresis losses are very low, these materials are most suitable for transformer cores and armatures, which are subjected to rapid reversals of magnetization.

SUMMARY

- **Crystalline solid:** A crystalline solid is one where the internal structure is regular in nature, the atoms within it being set in well-defined patterns.
- **Amorphous materials** do not possess repeated structures over a long range as in crystalline solids.
- Rubber is an example of a **polymer**. It consists of very long chain molecules.
- **Stress** is a quantity which defines the magnitude of forces that cause deformation. Stress is usually defined as force per unit area.
- When forces pull on an object and cause its elongation, like the stretching of an elastic band, we call such stress a **tensile stress**.
- **Strain** is given as a fractional change in either length (under tensile stress) or volume (under bulk stress) or geometry (under shear stress).
- The **tensile strength of a solid** is a measure of its resistance to being pulled apart.
- **Ductility** is a property of metal that enables it to be drawn through a die to produce a wire.
- **Elasticity** is a measure of a deformed object's ability to return to its original size and shape once the outside forces are removed.
- **Young's modulus** is defined as the ratio of stress to strain.
- **Bulk Modulus** If an object is subjected to inward forces from all sides, its volume will decrease.
- **Energy Band:** The electrons in the same orbit exhibit different energy levels. The grouping of these different energy levels is known as an energy band.
- **Superconductivity** allows charge to flow through a superconducting conductor without losing its energy to thermal energy.
- **Magnetic Hysteresis / Retentivity:** When a magnetic material is placed in an external magnetic field due to which its grains get oriented in its direction and the material is magnetized. If this material is removed from that external magnetic field, even some magnetization remains in that material which is called residual magnetism. This is also called retentivity of the material.
- **A magnetic domain** is a region within a magnetic material in which magnetization is in a uniform direction.
- **Curie point or Temperature** is defined as when some magnetic materials lose their magnetic property as temperature becomes above the Curie temperature.
- **Hysteresis Loss** is the disturbance in the alignment of the various domains that causes hysteresis loss.

TEXTBOOK WORK

SECTION- A: MULTIPLE CHOICE QUESTIONS (MCQS)

1. The property of a body, by virtue of which it tends to regain its original size and shape when the applied force is removed, is called. (2025)
(a) Elasticity (b) Plasticity (c) Rigidity (d) Compressibility
2. Substances which can be stretched to cause large strains are called.
(a) Brittle (b) Ductile (c) Plastic (d) Elastomer
3. If the load is increased beyond the point, the strain increases rapidly for even a small change in the stress.
(a) Elastic point (b) Yield point (c) Plastic point (d) Fracture point
4. The reciprocal of the bulk modulus is called.
(a) Compressibility (b) Volume stress (c) Modulus of rigidity (d) Volume strain
5. Which of the following statements is/are wrong?

- i. Hollow shaft is much stronger than a solid of the same length and same mass.
ii. Reciprocal of bulk modulus of elasticity is called compressibility.
iii. It is difficult to twist along rod as compared to small rod.
- (a) III only (b) I only (c) II and III (d) I and II
6. Metals are good conductors of heat and electricity. This property is conferred by. **(2024)**
(a) Covalent (b) Ionic (c) Metallic (d) Hydrogen
7. For a metallic crystal, delocalized electrons occupied band is: **(MP:25)**
(a) Conduction band (b) Valence band
(c) Conduction and valence bands (d) there are no delocalized electrons
8. The semiconductors have resistivity.
(a) Between conductor and insulator (b) More than insulators
(c) Less than conductors (d) Depending upon the semiconductor material property.
9. A ferromagnetic substance becomes a permanent magnet when it is placed in a magnetic field because
(a) All the domains get oriented in the direction of the magnetic field.
(b) All the domains get oriented in the direction opposite to the direction of the magnetic field.
(c) Domains get oriented randomly.
(d) Domains are not affected by the magnetic field.
10. Identify methods to demagnetize a ferrimagnet.
(a) By cooling, heating, or submerging in water
(b) By heating, hammering, and spinning it in an external magnetic field
(c) By hammering, heating, and rubbing with cloth
(d) By cooling, submerging in water, or rubbing with cloth

SECTION- B: SHORT ANSWERED QUESTIONS

1. Why are the springs made of steel and not of copper? **(2024)**
2. The breaking force for a wire is F . What will be the breaking force for two parallel wires of the same size?
3. Distinguish between intrinsic and extrinsic semiconductor.
4. A wire is replaced by another wire of same length and material but of twice the diameter. What will be the effect on the: (a) Increase in its length under a given load? (b) Maximum load which it can bear?
5. Sand does not possess any definite shape and volume, still it is solid. Give reason.
6. Specify the importance of stress-strain curves.
7. Why don't liquids possess rigidity?
8. Give applications of Curie point.
9. What are amorphous materials and what are their uses?

SECTION- C: LONG ANSWERED QUESTIONS

1. Explain Force-Extension graph.
2. Derive relation for Young's Modulus and Shear Modulus.
3. Distinguish between structure of crystalline, glassy, amorphous, and polymeric solids.
4. Describe the energy bands in solids.
5. Describe superconductivity and its applications.
6. Discuss the applications of superconductors for MRI, Maglev's, and supercomputers.
7. Describe hysteresis loss.
8. Synthesize hysteresis loops for relationship between magnetic field strength and magnetizing current.

9. Discuss energy bands and their classification. Explain magnetic properties of soft and hard magnetic materials.

SECTION- D: NUMERICAL

1. The 'lead' in pencils is a graphite composition with a Young's modulus of $1.0 \times 10^9 \text{ N/m}^2$. Calculate the change in length of the lead in an automatic pencil if you tap it straight into the pencil with a force of 4.0 N. The lead is 0.50 mm in diameter and 60 mm long. **(1.0mm)**
2. A wire of 2.2 m long and 2.25 mm in diameter, when stretched by a weight of 8.8 kg, its length has been increased by 0.25 mm. Find the stress, strain, and Young's modulus of the material of the wire. Given $g = 9.8 \text{ m/s}^2$ **($2.2 \times 10^7 \text{ N/m}^2$, 1.14×10^{-4} , $2 \times 10^{11} \text{ N/m}^2$)**
3. A farmer making juice fills a glass bottle to the brim and caps it tightly. The juice expands more than the glass when it warms up, in such a way that the volume increases by 0.2% (i.e., $\frac{\Delta V}{V_0} = 2 \times 10^{-3}$) relative to the space available. Calculate the normal force exerted by the juice per square centimeter, if its bulk modulus is $1.8 \times 10^9 \text{ N/m}^2$. Assuming that the bottle does not break. **(432 N/cm²)**
4. The elastic limit of copper is $1.5 \times 10^8 \text{ N/m}^2$. It is to be stretched by a load of 10 kg. Find the diameter of the wire if the elastic limit is not to be exceeded. **(0.912 mm)**
5. What would be the greatest length of a steel wire which is fixed at one end, and can it be hanged freely without breaking? The breaking stress of steel is $7.8 \times 10^8 \text{ N/m}^2$, and the density of steel is 7800 kg/m^3 . **($1.02 \times 10^4 \text{ m}$)**
6. A mild steel wire of radius 0.55 mm and length 3.5 m is stretched by a force of 52 N. Calculate: (a) Longitudinal stress, (b) Longitudinal strain, and (c) Elongation produced in the wire if Young's modulus is $2.1 \times 10^{11} \text{ N/m}^2$. **($5.47 \times 10^7 \text{ N/m}^2$, 2.6×10^{-4} , 0.91 mm)**
7. Calculate the change in volume of a lead block of volume 1.3 m^3 subjected to a pressure of 12 atm. Also, calculate the compressibility of lead. Given the bulk modulus as $B = 80 \times 10^9 \text{ N/m}^2$. **($1.97 \times 10^{-5} \text{ m}^3$, $1.25 \times 10^{-11} \text{ N/m}^2$)**
8. The thickness of a metal plate is 0.35 inches. It's drilled to have a hole of radius 0.08 inches on the plate. If the shear strength is $4 \times 10^4 \text{ lbs/in}^2$, determine the force needed to make that hole. **(0.176 in², $7 \times 10^3 \text{ lbs}$)**